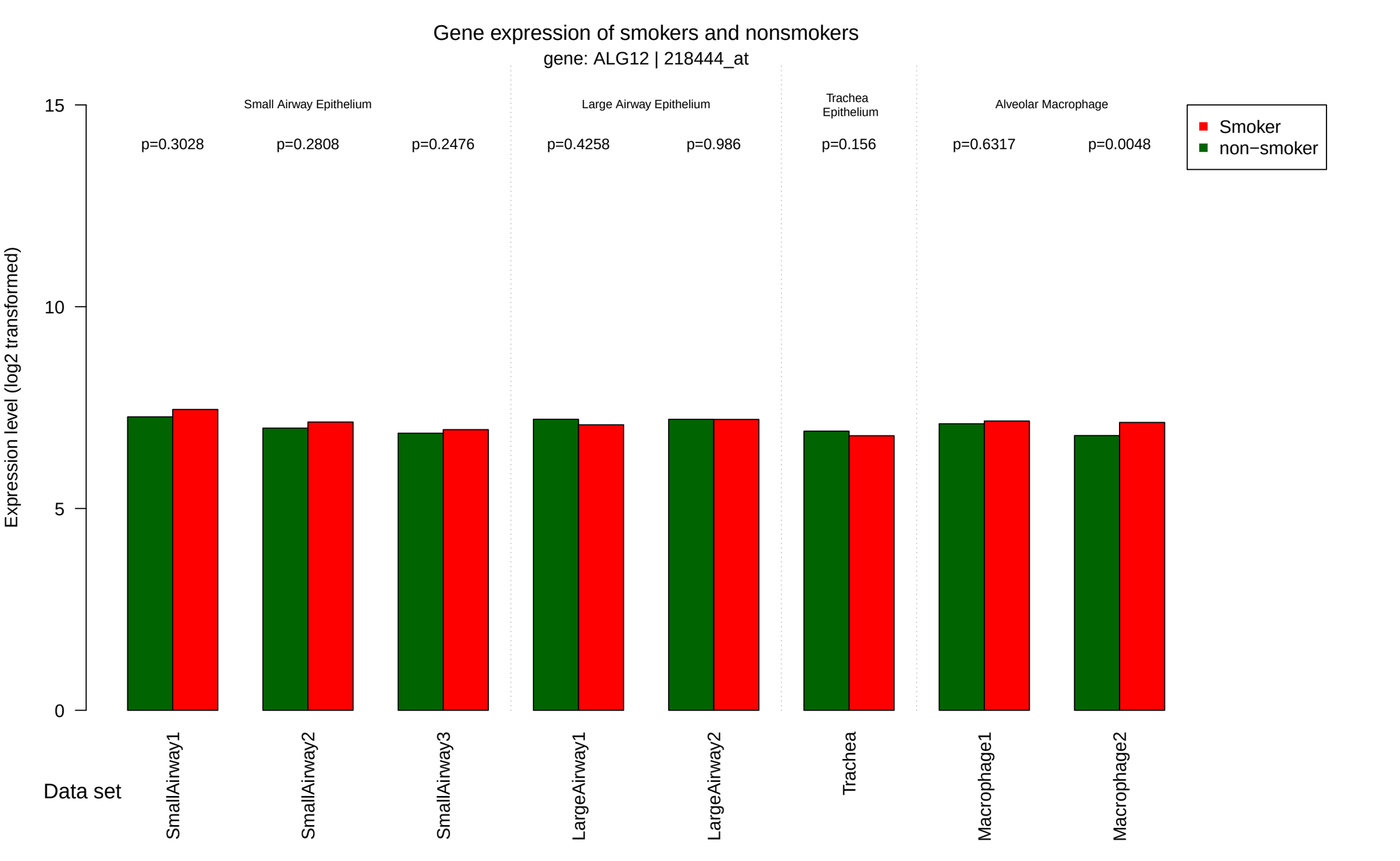
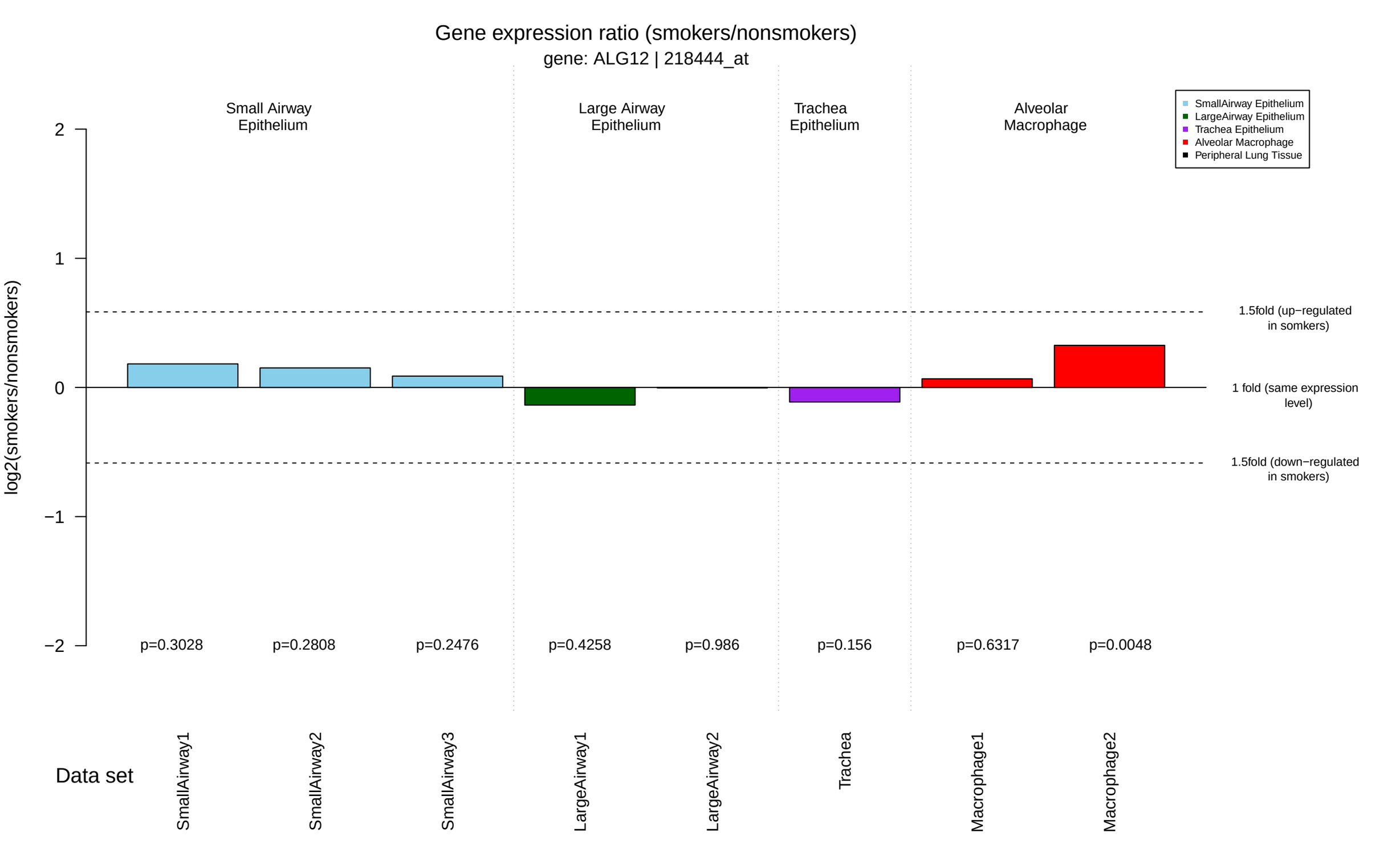
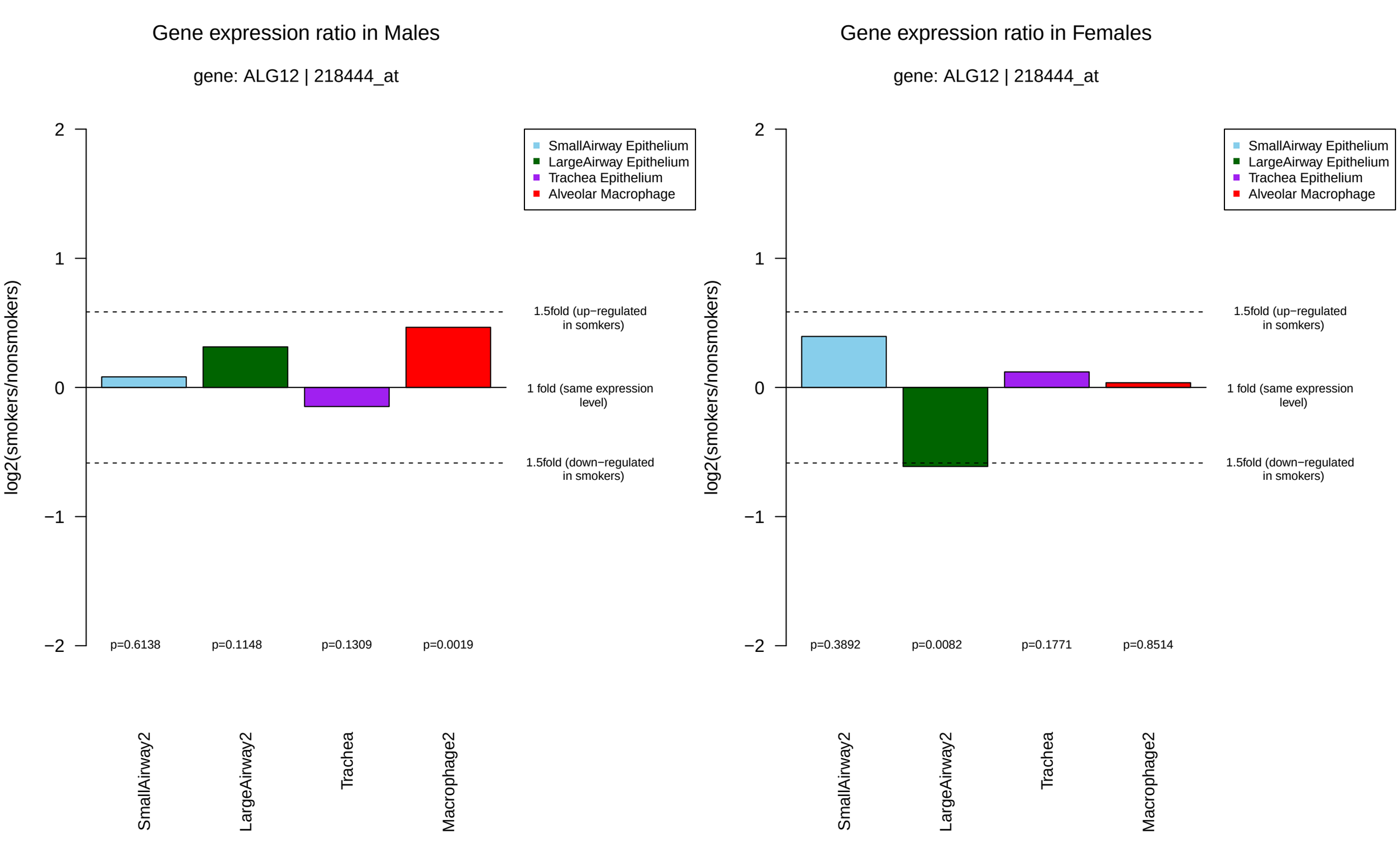
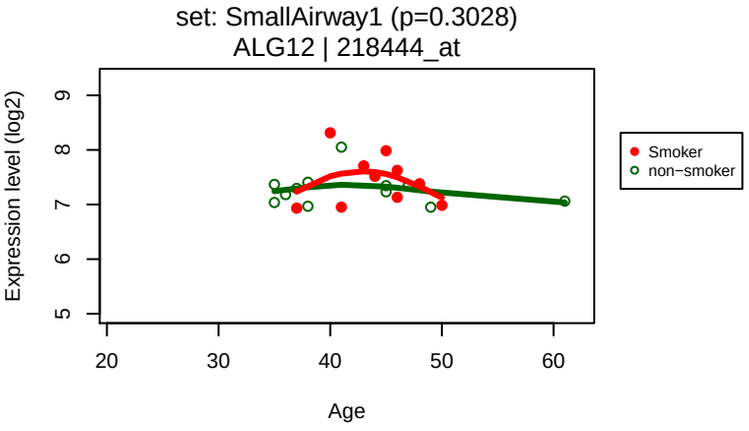
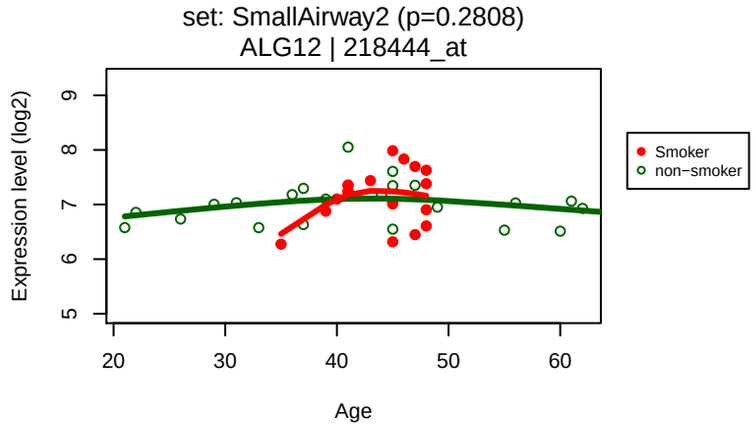
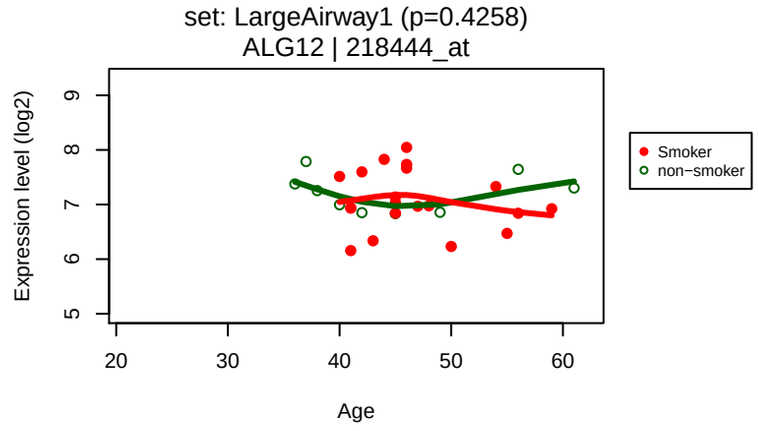

Gene expression of smokers and nonsmokers
gene: ALG12 | 218444_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3028
p=0.2808
p=0.2476
p=0.4258
p=0.986
p=0.156
p=0.6317
p=0.0048
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

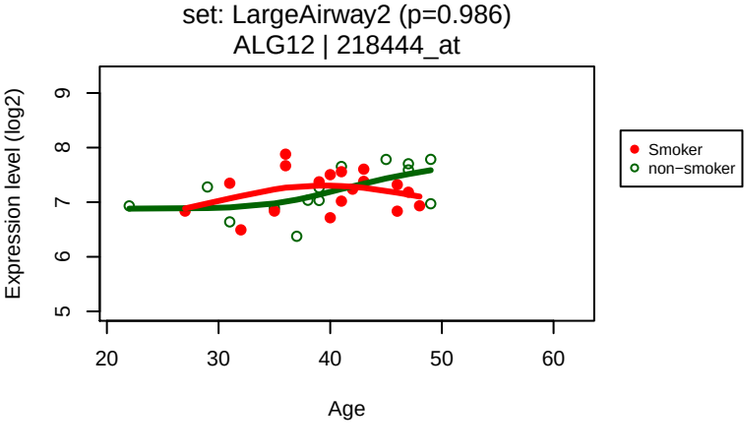

Gene expression ratio (smokers/nonsmokers)
gene: ALG12 | 218444_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3028
p=0.2808
p=0.2476
p=0.4258
p=0.986
p=0.156
p=0.6317
p=0.0048
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

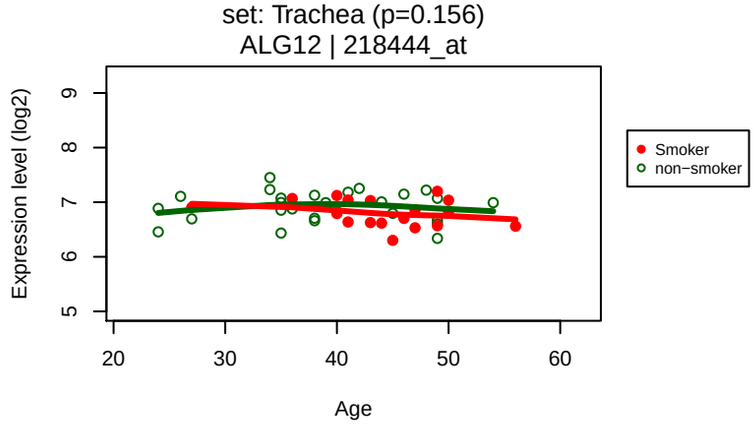

Gene expression ratio in Males
gene: ALG12 | 218444_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6138
p=0.1148
p=0.1309
p=0.0019
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALG12 | 218444_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3892
p=0.0082
p=0.1771
p=0.8514
SmallAirway2
LargeAirway2
Trachea
Macrophage2

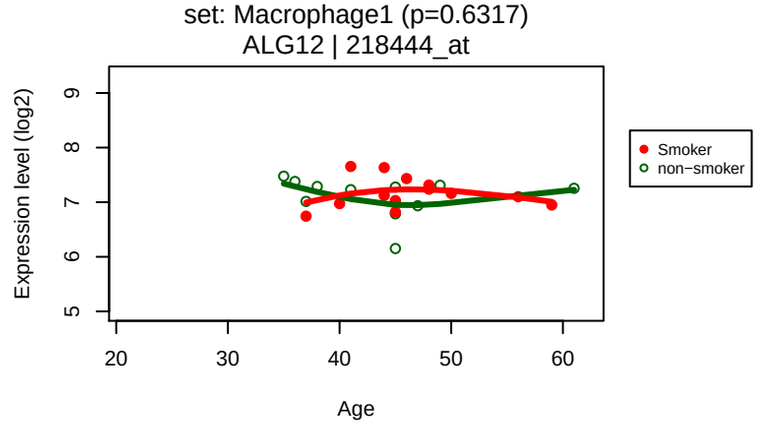

set: SmallAirway1 (p=0.3028)
ALG12 | 218444_at
Expression level (log2)
Age
Smoker
non−smoker

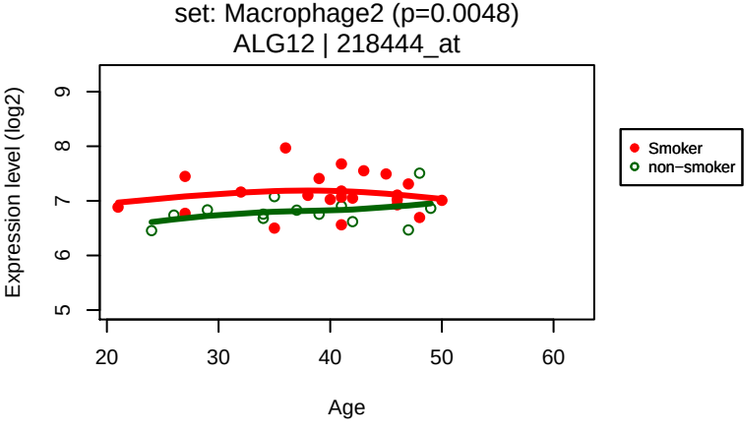

set: SmallAirway2 (p=0.2808)
ALG12 | 218444_at
Expression level (log2)
Age
Smoker
non−smoker

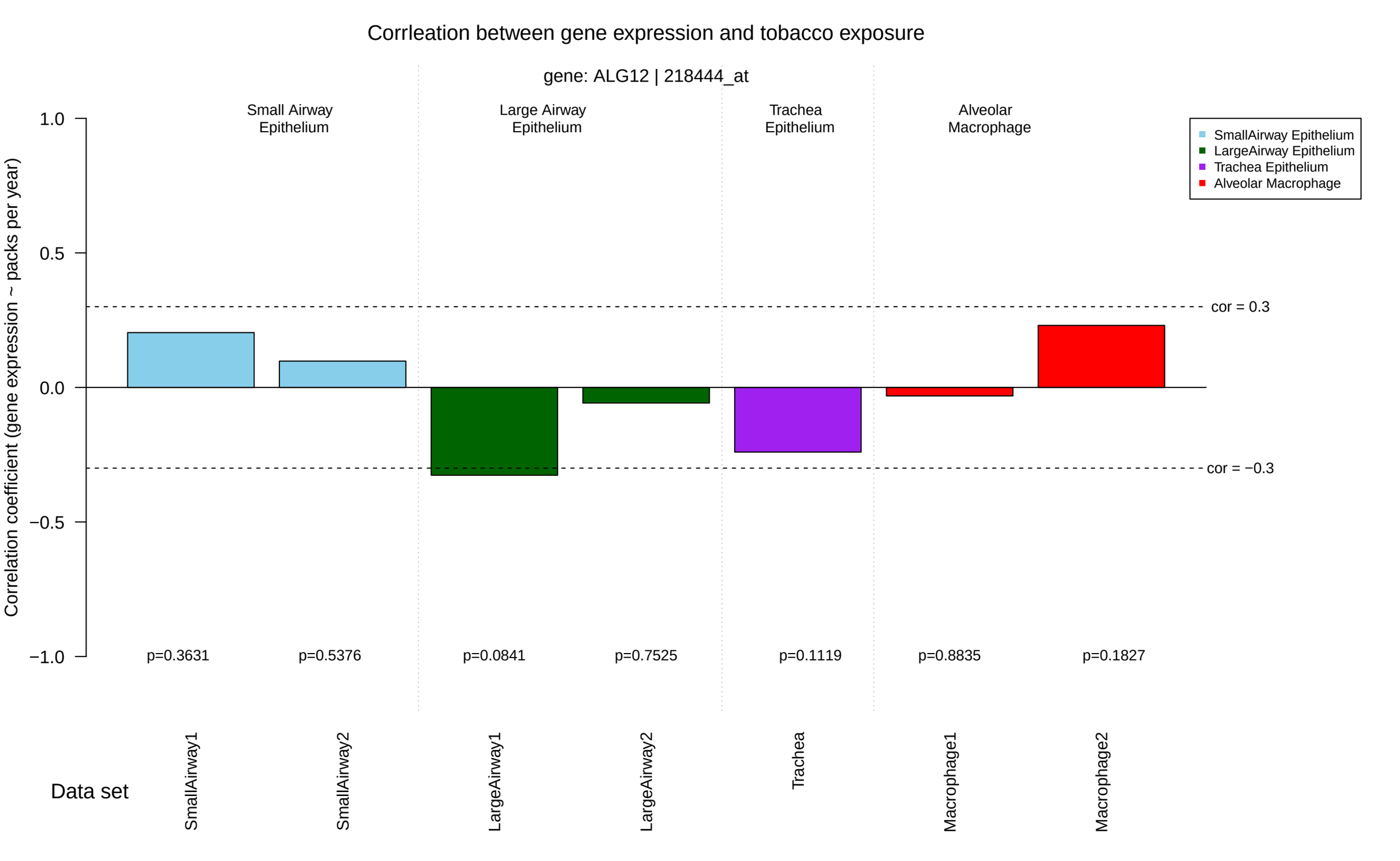

set: LargeAirway1 (p=0.4258)
ALG12 | 218444_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.986)
ALG12 | 218444_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.156)
ALG12 | 218444_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6317)
ALG12 | 218444_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0048)
ALG12 | 218444_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALG12 | 218444_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3631
p=0.5376
p=0.0841
p=0.7525
p=0.1119
p=0.8835
p=0.1827
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2